What Does EEG-to-fMRI Synthesis Add to EEG-Based Diagnosis? A Controlled Benchmark in Psychosis, Schizophrenia and Parkinson's Disease

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Abstract

Background: Electroencephalography (EEG) is inexpensive, whereas functional magnetic resonance imaging (fMRI) is spatially detailed but costly. EEG-to-fMRI synthesis has been proposed to support EEG-only diagnosis, but whether synthesized fMRI adds information beyond the EEG it is computed from has not been tested with appropriate controls.

Methods: We introduce a controlled evaluation protocol and apply it to four synthesizers, trained on simultaneous EEG-fMRI from healthy adults, and to harmonised resting EEG from five public cohorts: two first-episode psychosis samples (77 and 61 participants), chronic schizophrenia (77), Parkinson's disease (50) and a task-EEG schizophrenia-spectrum sample (43). Synthesized fMRI, an EEG band-power baseline and their combination are compared under nested repeated cross-validation, permutation tests and paired bootstrap, with untrained (random-weight) synthesizers as an architecture control.

Results: Band power separated Parkinson's disease from controls (area under the ROC curve, AUC, 0.79) but not the psychosis or schizophrenia cohorts (0.54–0.65). Synthesized fMRI classified Parkinson's disease (best AUC 0.74) but stayed below band power; adding it to band power gave no gain that survived correction for multiple comparisons; untrained synthesizers reproduced the Parkinson's

disease result (0.70–0.74); and contrastive fine-tuning reached at most 0.70. The synthesizers did not outperform an EEG-free template image, and EEG explained a negligible part of the blood-oxygen-level-dependent signal (correlation 0.03). **Conclusions:** Synthesized fMRI reflected EEG spectra, largely through the network architecture, and added no reliable diagnostic information beyond band power. The protocol, code and checklist of controls set a baseline for future diagnostic claims.

Keywords: Electroencephalography, EEG-to-fMRI synthesis, Functional MRI, Schizophrenia, First-episode psychosis, Parkinson's disease, Diagnostic classification, Cross-modal generation, Benchmark, Evaluation protocol

1 Introduction

fMRI resolves brain activity in space but is costly and scarce, especially in low-resource settings Ogbole et al. (2018); EEG is cheap and portable but spatially coarse. EEG-to-fMRI synthesis learns, from simultaneous recordings, a mapping from scalp EEG to BOLD volumes, and has been proposed as a way to obtain fMRI-like information where only EEG is available Liu et al. (2019, 2020); Bricman et al. (2021); Calhas et al. (2023); Roos et al. (2025); He et al. (2025); Lanzino et al. (2024); Li et al. (2024). Schizophrenia and related psychotic disorders are a natural target: they are clinically heterogeneous Saha et al. (2005), their diagnosis draws on psychological, neuroimaging, molecular and speech-based assessments Andreou et al. (2020); Neale et al. (2015); Corcoran et al. (2020), and both EEG TAN et al. (2016); Aydın et al. (2018); Roach et al. (2008); Bates et al. (2009) and fMRI Oh et al. (2020); Velakoulis et al. (2006); Haukvik et al. (2013) carry group differences. Parkinson's disease, with well-documented resting EEG alterations, offers a second target. Two studies have classified patients from synthesized fMRI: schizophrenia from a task-EEG cohort, in an earlier version of this work Calhas and Henriques (2023), and Parkinson's disease from eight subjects of a resting-EEG cohort Yao et al. (2025).

Whether synthesized fMRI adds diagnostic information beyond the EEG it is computed from has not been tested with the controls such a claim requires: classification

data without acquisition differences between groups; a comparison, under identical validation, with an EEG-only view and with the combination of both; and a test of whether any discrimination comes from the learned EEG-to-fMRI mapping or from the network architecture alone, which can be done by applying the same networks with untrained weights. We report such an evaluation across four synthesizers and five clinical cohorts.

Contributions. (1) An evaluation protocol for diagnostic claims from synthesized fMRI: a harmonisation pipeline that maps any clinical EEG onto the synthesizers' training montage and removes non-EEG channels by type; an EEG-only baseline and a fusion test under identical nested validation; untrained-network controls; and subject-level permutation and paired bootstrap inference. (2) A benchmark applying this protocol to four synthesizers (CNN-TAG, E2fNet, Spec2VolCAMU-Net and NT-ViT) and to a contrastive fine-tuning method Calhas and Henriques (2023) on five public cohorts spanning first-episode psychosis, chronic schizophrenia, a schizophrenia-spectrum task cohort and Parkinson's disease. (3) Supporting evidence on what the synthesizers learn: a scanner-trigger-aligned training protocol for the NODDI dataset, an EEG-free template baseline and a fair-target test of how much BOLD variance EEG predicts. (4) A checklist of controls for diagnostic uses of synthesized fMRI (Section 5), with open code. In brief, the synthesized volumes carried the EEG spectral information they are computed from, largely through the network architecture, and added no reliable diagnostic information beyond EEG band power in these cohorts.

2 Related work

EEG-to-fMRI synthesis. Convolutional transcoders Liu et al. (2019, 2020), attention- and Fourier-feature-based encoder-decoders Calhas et al. (2023), U-Net and Mamba decoders Roos et al. (2025); He et al. (2025), vision transformers Lanzino et al. (2024) and diffusion models Yao et al. (2025) synthesize volumes, while NeuroBOLT

and UnEBOLT Li et al. (2024, 2026) and distributed-lag models Mehta et al. (2025) predict region time series; the latter work showed, within subjects and with surrogate tests, that EEG predicts BOLD in task-relevant regions. Downstream diagnostic use has been reported for schizophrenia Calhas and Henriques (2023) and Parkinson's disease Yao et al. (2025).

Schizophrenia. Schizophrenia is marked by cognitive and structural brain alterations Sui et al. (2018). fMRI-based classification of controls and patients has shown promise de Filippis et al. (2019), with consistent findings of frontal and temporal volume reductions Chin et al. (2018) and white-matter abnormalities Cetin-Karayumak et al. (2020). EEG-based classification is common, but its pitfalls, including small samples and validation that is not at the subject level, have been documented Barros et al. (2021); acquisition differences between groups in public releases are a further risk.

Interpretability. For EEG, interpretability remains a challenge. Pinto et al. (2022) applied evolutionary feature selection and *apriori* association rules to explain seizure detection, identifying key features across spectral, temporal, and electrode dimensions. Band power, used here as the EEG baseline, is itself spatially organised over the scalp and has established clinical interpretations in schizophrenia Roach et al. (2008); Bates et al. (2009); the case for synthesized fMRI as a more interpretable view would require spatial maps validated against real fMRI in patients, which the present data cannot provide.

3 Materials and methods

Figure 1 summarises the study design.

3.1 Datasets

Table 1 summarises the datasets. All are previously published, anonymised public releases analysed here as secondary data.

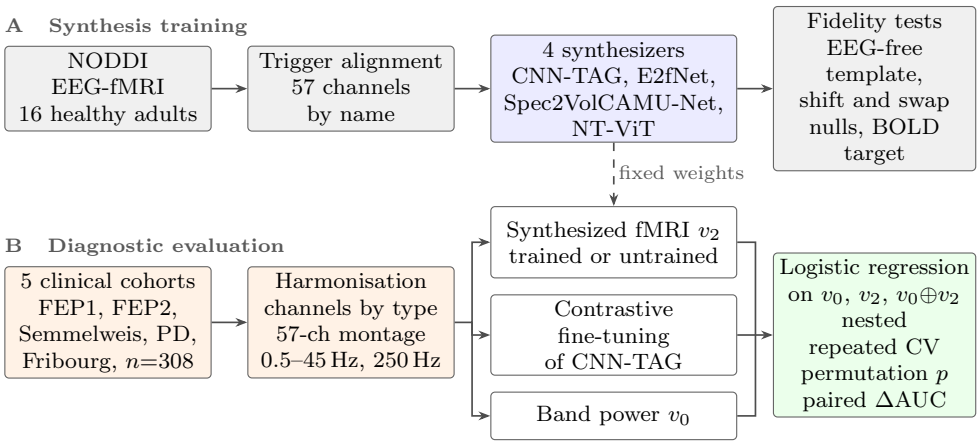


Fig. 1 Study design. Four synthesizers are trained on simultaneous EEG-fMRI of healthy adults (NODDI) with scanner-trigger alignment and kept fixed. Resting EEG of five clinical cohorts is harmonised to the training montage and represented by band power (v_0), by the synthesized fMRI of each synthesizer with learned or random weights (v_2), and by their combination ($v_0 \oplus v_2$); a contrastive fine-tuning of CNN-TAG is evaluated alongside. Every view is classified with the same nested cross-validation and permutation tests. Supporting analyses on NODDI test what the synthesizers learn.

First-episode psychosis (FEP1, FEP2). Two independently collected samples from the University of Pittsburgh (OpenNeuro ds003944 and ds003947) Salisbury et al. (2021a,b): individuals with a first episode of psychosis and healthy controls; five minutes of eyes-open resting EEG recorded with a 10-10 cap inside an Elekta Neuromag Vectorview MEG system (61 EEG channels, 1000 Hz, reference TP9). Age, sex, symptom and functioning scales (BPRS, SANS, SAPS, GAF, SFS), cognition (MATRICS, WASI) and chlorpromazine-equivalent medication are released. FEP1 lists 82 recordings, but five people appear twice under the identifiers `sub-X` and `sub-XA` (same age, sex and race, recorded days apart); we kept one recording per person (47 patients, 30 controls). FEP2 contains 31 patients and 30 controls and excludes, by construction, the subjects shared with FEP1.

Chronic schizophrenia (Semmelweis). Resting EEG from Semmelweis University, Budapest Racz et al. (2025): 38 patients with schizophrenia and 39 age- and sex-matched controls; about two minutes of eyes-closed rest, 64-channel Neuroscan amplifier at 1000 Hz. Patients are identified by the presence of PANSS ratings, which

reproduces the 38/39 split of the source publication (35 of the 38 carry an ICD-10 F20/F25 code in the released table; three have a missing or non-F2 code). Medication, illness duration, education and PANSS scores are released.

Parkinson's disease (PD). Resting EEG from the University of New Mexico (OpenNeuro ds003490) Cavanagh (2021): 25 patients with Parkinson's disease and 25 matched controls; 63 EEG channels, 500 Hz (Brain Vision ActiChamp; reference CPz), instructed one-minute eyes-closed and eyes-open spans, plus EOG and a three-axis accelerometer on the most tremor-affected hand. Patients were recorded twice, a week apart, ON and OFF dopaminergic medication (13 ON first, 12 OFF first); we analysed the OFF session. Eight subjects of this cohort were used by Yao et al. (2025).

Schizophrenia spectrum, task EEG (Fribourg). The cohort used in Calhas and Henriques (2023), released by Padée et al. (2022): 19 patients with a psychotic disorder (ICD-10) and 24 controls, recorded during the Ultimatum Game (responder role), 128-channel Biosemi ActiveTwo at 2048 Hz; the release includes additional EXG channels whose layout differs between groups (four in all controls, six in 18 of 19 patients), which are excluded by the channel selection of Section 3.3. The public release distributes the group label only; group-level characteristics are reported in Horat et al. (2018).

Data checks. For every cohort we checked for repeated participants, verified group sizes against the source publications, and tested whether recording metadata (channel set, sampling rate, filters, duration, recording date and hour, file size) differs between groups; fixed-length crops remove duration differences.

3.2 Simultaneous EEG-fMRI for synthesis (NODDI)

fMRI is available only in the synthesis training dataset, NODDI Deligianni et al. (2014, 2016); the clinical cohorts are EEG-only and contain no fMRI. In NODDI, fMRI was acquired simultaneously with EEG at rest (eyes open, fixation) on a 1.5 T Siemens Avanto: T2*-weighted gradient-echo EPI, TR/TE 2160/30 ms, 30 slices of 3 mm (1 mm

Table 1 Characteristics and roles of the datasets. *pt*: patients; *hc*: healthy controls; age as mean ± standard deviation (years); analysed EEG is the fixed crop used for every subject of a cohort.

Dataset	Role / disease	pt/hc	Age pt — hc	Sex (M/F) pt — hc	Recording (analysed EEG)
NODDI	synthesis training	0/16	32.8 ± 8.1[a]	11/6[a]	rest, eyes open, simultaneous EEG-fMRI, 64 ch, 250 Hz
FEP1 (ds003944)	first-episode psychosis	47/30	22.5 ± 5.1 — 21.6 ± 4.1	33/14 — 18/12	rest, eyes open, 61 ch, 1 kHz (300 s)
FEP2 (ds003947)	first-episode psychosis	31/30	23.4 ± 4.7 — 24.2 ± 5.1	22/9 — 16/14	rest, eyes open, 61 ch, 1 kHz (300 s)
Semmelweis	schizophrenia (chronic)	38/39	33.9 ± 10.6 — 32.7 ± 9.7	24/14 — 24/15	rest, eyes closed, 64 ch, 1 kHz (118 s[b])
PD (ds003490)	Parkinson's disease (OFF)	25/25	69.7 ± 8.7 — 69.3 ± 9.6	16/9 — 16/9	rest, eyes closed/open spans, 63 ch, 500 Hz (400 s)
Fribourg (ds004000)	schizophrenia spectrum	19/24	n/a	n/a	Ultimatum Game, 128 ch, 2048 Hz (400 s)

[a] For the 17 acquired participants; version 2 of the release contains 16. [b] One control recording is 109.7 s long and was used in full.

gap), effective voxel size $3.3 \times 3.3 \times 4.0$ mm, $64 \times 64 \times 30$ voxels, 300 volumes; EEG with a 64-channel MR-compatible cap (BrainCap MR), recorded at 5 kHz, released at 250 Hz after artefact correction.

Two properties of this release matter for synthesis and, to our knowledge, have not been reported. First, in 15 of the 16 subjects the EEG begins 13–130 s before the first fMRI volume; pairing EEG and fMRI from the start of the file would pair EEG with volumes 6–60 TRs away. We pair them through the scanner volume markers in the EEG file (one per TR). Second, two electrode montages are present: 10 subjects have F1/F2/C1/C2/P1/P2 and 6 have P9/P10/PO9/PO10/O9/O10, in a different channel order, so positional channel indexing mixes electrodes across subjects. We use the 57 channels recorded in all subjects, selected by name; the ECG channel is excluded.

3.3 EEG harmonisation and preprocessing

EEG datasets differ in electrode count and layout, sampling rate, reference and paradigm, which changes their representation in $\mathbb{R}^{C\times T}$. We apply a single harmonisation to every recording, clinical or not, so that each synthesizer receives inputs in exactly the montage and sampling rate it was trained on:

Channel selection by type. EEG channels are selected by their declared type, never by position; EOG, ECG, EMG, accelerometer, trigger and auxiliary (EXG) channels are removed. Selection by type also keeps acquisition differences unrelated to the disease out of the input: in the Fribourg release, for example, the auxiliary (EXG) channel layout differs between patients and controls.

Montage harmonisation. Channels are mapped to standard 10-05 positions and interpolated with spherical splines onto the 57 EEG channels recorded in every NODDI subject (Section 3.2).

Common filtering and resampling. Notch filtering at the mains frequency and harmonics, 0.5–45 Hz band-pass, average reference and resampling to 250 Hz. Frequencies above 45 Hz are discarded for every dataset, so all synthesizers see the same frequency range.

Clinical EEG has no fMRI volume markers; the synthesizers are applied to consecutive pseudo-TRs of 2.16 s starting 43.2 s into each recording, so that every synthesized volume has a full 20-segment history.

All recordings went through this harmonisation (MNE-Python). Each clinical recording was cropped to a fixed duration per cohort (Table 1), identical for patients and controls, to remove duration differences between groups. The cleaned NODDI EEG released by the dataset authors, in which scanner (gradient) and pulse (ballistocardiographic) artefacts were removed Deligianni et al. (2014), was used as is before harmonisation.

3.4 Problem formulation and EEG views

Let $\mathcal{V} = \{v_1, \ldots, v_n\}$ be a set of different views, such that $\forall i \in \{1, \ldots, n\} : v_i = \{F_i, \theta_i\}$. A view v_i is characterized by a function structure, F_i, and its parameters, θ_i. Each projection $F_i : \mathbb{R}^{D_0} \rightarrow \mathbb{R}^{D_i}$ is performed from the original feature space, $\mathbb{R}^{D_0}$, to its view space, $\mathbb{R}^{D_i}$. Each subject's recording, $\vec{x}_0$, is paired with a label $y \in \{0, 1\}^K : \sum_k y_k = 1$, where $K = 2$ (patient or control). A view, v_i, is evaluated by a linear classifier minimizing

$$\mathcal{L}_C(\vec{x}_0, y) = -y \times \log(\sigma(W_f^i \cdot v_i(\vec{x}_0) + b_f^i)), \tag{1}$$

where $W_f^i \in \mathbb{R}^{K \times D_i}$ and $b_f^i \in \mathbb{R}^K$ are the parameters of the linear classifier, referred to as $\theta_C^i = W_f^i \bigcup b_f^i$, using the softmax activation, σ, plus an l_2 penalty on W_f^i. The question is whether a synthesized *fmri* view, alone or added to an EEG view, lowers the expected loss on unseen subjects relative to the EEG view alone.

3.4.1 *bandpower* view

Frequency-domain EEG features are long-established markers in schizophrenia and Parkinson's disease Itil (1977); Roach et al. (2008); Bates et al. (2009); Barros et al. (2021). We use the simplest such view: for each of the $C = 57$ channels (Section 3.3), Welch power spectra (4 s windows) summarised as relative log power in five bands (1–4, 4–8, 8–13, 13–30, 30–45 Hz), so $v_0(\vec{x}_0) \in \mathbb{R}^{C \times 5}$. Band power is spatially organised over the scalp and clinically interpretable, so it is both the natural EEG-only baseline and the information any synthesized view starts from.

3.4.2 *stft* view

Research studies, along the years, have had a common trend of extracting time frequency features from the EEG signal Fingelkurts et al. (2007); Burgess et al. (2002);

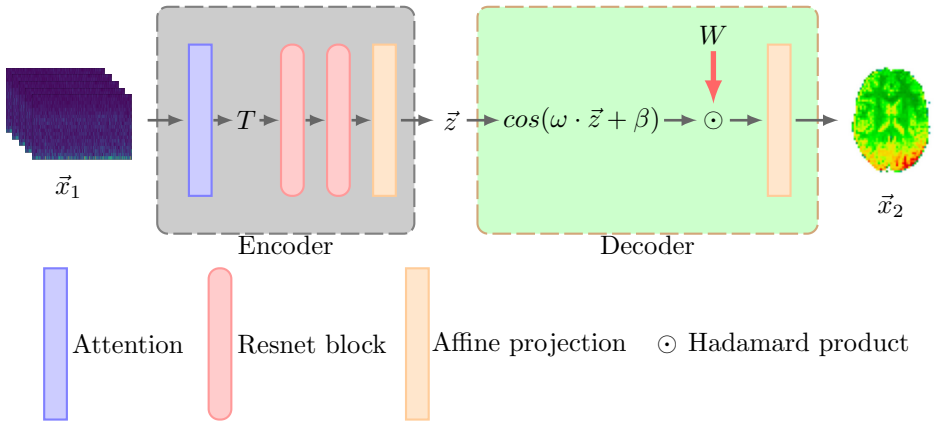


Fig. 2 Architecture of the CNN-TAG synthesizer. The neural architecture has two components: an Encoder (shaded in grey) and a Decoder (shaded in green). The input is the *stft* v_1 representation. The output is the synthesized *fmri*. The Encoder begins with a simple attention mechanism on the channels dimension of the *stft*. After, it is processed by two Resnet blocks and an affine layer. This produces the latent representation $\vec{z}$. Following, the Decoder picks this representation and builds the cosine bases through the projection $\omega \cdot \vec{z} + \beta$. The sinusoids are style induced with $cos(\omega \cdot \vec{z} + \beta) \odot W$. The vector W is a style fixed pretrained vector of an *fmri* representation, learned from a simultaneous EEG and fMRI dataset (described in Section 3.2). Finally, an affine layer projects it to the *fmri* space.

Albrecht et al. (2018); Roach et al. (2008); Bates et al. (2009). The synthesizers take a short-time Fourier representation Jacobsen et al. (2003) as input, $v_1 = \{F_1, \theta_1\}$ with $\vec{x}_1 \in \mathbb{R}^{D_1}$, $D_1 = C \times F \times T$: log-magnitude spectra of $T = 20$ consecutive EEG segments of one fMRI repetition time (TR, 2.16 s) each, between 0.46 and 45 Hz ($F = 97$ bins), i.e. the 43.2 s of EEG preceding an fMRI volume, which covers the haemodynamic delay. This view is not classified directly; it defines the input of the *fmri* views.

3.5 Synthesizers and the synthesized fMRI view

The first synthesizer is the network introduced and validated in our earlier work Calhas et al. (2023); its architecture is adopted from that study, not proposed here, and the present work re-implements it in PyTorch and retrains it with the protocol of Section 3.2. We refer to it as CNN-TAG. Its flow is illustrated in Figure 2. Its input is the *stft* view, $\vec{x}_1$; its output is the *fmri* view, $\vec{x}_2 \in \mathbb{R}^{M_1 \times M_2 \times M_3}$, a volume of $64 \times 64 \times 30$ voxels. Zooming out of the neural network, we describe this view as

$v_2 = \{F_2, \theta_2\}$, where $F_2 = F_1 \bigcup F$ is the union of the *stft* transform and the network F, and $\theta_2 = \theta_1 \bigcup \theta_F$.

In terms of structure, the network processes the input with an attention mechanism over channels. An attention weight matrix $A \in \mathbb{R}^{C\times(FT)}$ is used to compute $E \in \mathbb{R}^{C\times C}$, $E_{ij} = \vec{x}_i^{\top} A_j$, which is normalised with sparsemax along j to give B; the attention-processed representation is

$$T_i = \sum_{j}^{C} b_{ij}\, \vec{x}_j, \forall i \in \{1, \ldots, C\}. \tag{2}$$

T is processed by two residual blocks He et al. (2016) and a fully connected layer, whose parameters θ_E form the encoder. The encoder output projects random Fourier features Rahimi et al. (2007); Tancik et al. (2020), parameterized by ω and β and layer-normalised around $\pi/2$; the resulting sinusoids are multiplied with a learned latent style vector, $W \in \mathbb{R}^L$, and an affine transformation maps them to the fMRI volume space. With this, we formulate the function structure $F : \mathbb{R}^{C\times F\times T} \to \mathbb{R}^{D_2}$.

Other synthesizers. Three published synthesizers were trained with the same data and protocol, using the authors' model code: **E2fNet** Roos et al. (2025), a convolutional EEG encoder followed by a 2D U-Net and a convolutional fMRI decoder; **Spec2VolCAMU-Net** He et al. (2025), a multi-scale convolutional spectrogram encoder with temporal self-attention and a Vision-Mamba U-Net decoder, whose selective scan we computed with an exact pure-PyTorch chunked recurrence because the CUDA kernel was unavailable; and **NT-ViT** Lanzino et al. (2024), a mel-spectrogram vision-transformer encoder and decoder with a domain-matching loss, which computes its own spectrograms and therefore received the same 43.2 s window as raw waveforms (the published model used a simultaneous 2.16 s window without haemodynamic lag).

CATD Yao et al. (2025) could not be included because no code is available; NeuroBOLT and UnEBOLT Li et al. (2024, 2026) predict parcel time series rather than volumes and were used in the supporting fidelity analyses only.

Untrained controls. For each synthesizer, the same architecture with random initial weights was passed through the identical pipeline. A fixed nonlinear projection of EEG spectra into a voxel grid can support classification on its own; this control separates that effect from the learned EEG-to-fMRI mapping.

Synthesizer training. The synthesis target is the raw EPI volume min–max scaled to $[0, 1]$ per volume, the convention of the published volumetric synthesizers. Synthesizers used for classification were trained on 14 NODDI subjects, with two subjects held out to select the training epoch; loss $0.5(1 - \mathrm{SSIM}) + 0.5\,\mathrm{MSE}$, Adam, learning rate 10^{-3}, batch size 32, mini-batches shuffled across subjects, 50 epochs (15 for Spec2VolCAMU-Net). NT-ViT used the authors' recipe (AdamW, learning rate 5×10^{-4}, 30 epochs) with batch size 32 instead of 128, because the 43.2 s input did not fit in memory at 128. Input features were standardised with training-subject statistics only. NODDI carries no diagnostic labels and is never used for classification.

From synthesized volumes to a subject view. For each clinical subject, the synthesized volumes (one per pseudo-TR, clipped to $[0, 1]$) are summarised by their voxelwise temporal mean and standard deviation, giving $2 \times 64 \times 64 \times 30$ features; this is the *fmri* view classified in Section 3.7.

3.6 Contrastive sinusoid separation

In addition to the four synthesizers, we evaluate the integration approach of the earlier version of this work, which adapts the CNN-TAG synthesizer to the diagnostic task by separating classes in its sinusoid space while the voxel decoder is kept fixed.

CNN-TAG passes its encoder output $\vec{z}$ through random Fourier features, $\cos(\omega \cdot \vec{z} + \beta)$. Because the cosine is periodic, classes separated in $\vec{z}$ can overlap after the

projection. The method therefore layer-normalises Ba et al. (2016) $\vec{z}$ around $\pi/2$ with standard deviation $\pi/2$, which places most projections in an interval where the cosine is monotonic, and trains the encoder with the contrastive loss

$$\mathcal{L}_D(p_1, p_2, y_p) = y_p \times D(p_1, p_2) + (1 - y_p) \times ||D(p_1, p_2) - m||_1, \tag{3}$$

where p_1, p_2 formulate a pair of two instances and $y_p \in \{0, 1\}$ defines the pairwise label, being 0 when the label of p_1, p_2 mismatch and 1 when they match. $D(p_1, p_2)$ is the $l1$ distance $||p_1 - p_2||_1^1$. Setting $m \in [\epsilon, \pi] : \epsilon > 0$, together with the described layer normalization, allows separation of the data within a non periodic sub domain of the cosine. Under the random projection, the expected similarity of two projected points decreases exponentially with their latent distance, so a margin that separates the classes in $\vec{z}$ also separates them after projection; Additional file 1 gives the statement, its proof and an illustration of the periodicity problem. Figure 3 illustrates the effect before and after training.

Training. The encoder is trained with $\mathcal{L}_D$ on pairs of EEG windows, while a linear classifier with an l_1 penalty is trained with $\mathcal{L}_C$ on the synthesized volume (detached from the encoder); the style vector and the voxel decoder stay frozen at their NODDI weights. Each synthesized window is an instance, and a subject's score is the mean predicted probability over its windows. We run two variants: *re-initialised*, in which the encoder and Fourier projection start from random weights while the decoder keeps its NODDI weights, as in the original formulation, and *pretrained*, in which they start from the NODDI-trained synthesizer and are fine-tuned. Training uses a fixed budget (10 epochs, Adam, learning rate 10^{-3}, batches of 16 pairs, l_1 weight 10^{-4}), without per-fold tuning.

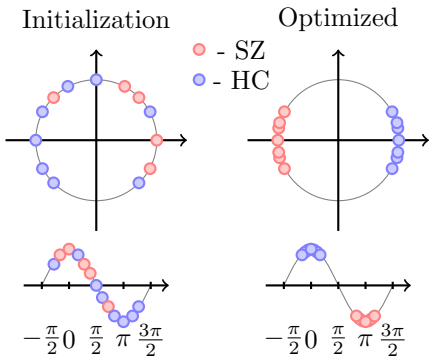


Fig. 3 Contrastive separation of sinusoids. Normalization of data points inside the unit circle, using layer normalization, along with the optimization of a contrastive loss lead to correct separation of sinusoids. Data points belong to two classes, HC and SZ, that are separated after the minimization of $\mathcal{L}_D$. Because we separate false pairs, according to $(1-y_p) \times ||D(x_1, x_2) - m||_1$, all points are placed within a shift variant interval of the cosine. The variance needed for classification.

3.7 Classification and statistical analysis

The experiments compare, per cohort, the following views, each classified with an l_2-regularised logistic regression (Equation 1), a linear classifier that allows a direct assessment of representational discriminative power and easy interpretation Treviso et al. (2020): the *bandpower* view v_0; the *fmri* view v_2 of each synthesizer, reduced to 20 principal components; their combination $v_0 \oplus v_2$; the same *fmri* and combined views from the untrained synthesizers; and the contrastive sinusoid-separation classifier of Section 3.6 (two variants). All transformations (standardisation, principal components) are fit on training folds only. All experiments were run on NVIDIA A100, H100 and L4 GPUs (synthesis) and CPU nodes (linear models).

3.7.1 Validation

To validate each view v_i, we perform repeated stratified 10-fold cross-validation at the subject level (20 repeats; 3 for the contrastive classifier), with one recording per person, and select the regularisation strength $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ by inner stratified 5-fold cross-validation within each training fold (Algorithm 1). We report the AUC averaged over repeats with a 95% bootstrap confidence interval

Algorithm 1 Nested repeated cross-validation for view v_i

Require: individuals S with labels y; view v_i; hyperparameter grid $\mathcal{H}$; R repeats
Ensure: AUC, out-of-fold probabilities P

for $r = 1$ **to** R **do**
  for each stratified fold $(S_{\text{train}}, S_{\text{test}})$ of S **do** ▷ 10 folds, subject level
    fit standardisation (and PCA) on S_{train} only
    $h^\star \leftarrow \arg\max_{h \in \mathcal{H}} \mathrm{CV}_5(v_i, S_{\text{train}}, h)$ ▷ inner folds within S_{train}
    fit v_i with $h^\star$ on S_{train}; $P_r[S_{\text{test}}] \leftarrow v_i.\textsc{predict}(S_{\text{test}})$
  end for
end for
return $\frac{1}{R}\sum_r \mathrm{AUC}(P_r, y)$, $\frac{1}{R}\sum_r P_r$

(2,000 stratified resamples of subjects), and a one-sided label-permutation p-value, $p = (1 + \#\{\mathrm{AUC}_{\text{perm}} \geq \mathrm{AUC}\})/(1 + N)$, from $N = 100$ permutations that each re-run the full nested procedure ($N = 20$ for the contrastive classifier, minimum $p \approx 0.048$). The added value of synthesis is $\Delta\mathrm{AUC} = \mathrm{AUC}(v_0 \oplus v_2) - \mathrm{AUC}(v_0)$, with a paired bootstrap interval over subjects. Leave-one-subject-out estimates are reported only as a sensitivity analysis: with weak features they are biased below chance, because the class balance of each training set mirrors the held-out label Varoquaux (2018).

3.8 Supporting synthesis analyses

To interpret the classification results we characterise what the synthesizers learn on NODDI. (a) *Image protocol*: leave-one-subject-out over the 16 subjects (two validation subjects per fold for model selection), scored with SSIM, RMSE and voxelwise temporal correlation against an EEG-free population template (the mean training volume), with trigger-aligned EEG, EEG paired from the file start, and ECG only. (b) *Fair target*: BOLD motion-corrected (rigid registration, FSL), cleaned with the Friston-24 motion model and spike regressors for framewise displacement above 0.5 mm Friston et al. (1996); Power et al. (2014), high-passed at 0.01 Hz, z-scored, smoothed (6 mm) and averaged into the 48 Harvard-Oxford grey-matter parcels present in every subject plus the global signal; ridge regression from lagged band power or full spectra, and

UnEBOLT Li et al. (2026) with its pretrained EEG encoder Jiang et al. (2024), predicted the parcel time series of held-out subjects. Significance used circular-shift and cross-subject swap nulls on detrended predictions and targets, and IAAFT surrogates of the BOLD with model refitting Schreiber and Schmitz (1996); Mehta et al. (2025).

3.9 Ethics approval and use of language models

This study is a secondary analysis of anonymised, publicly released data; no new data were collected. All original studies were approved by the responsible committees and obtained informed consent from participants, as reported by the data providers: NODDI, approval ID 4290/001 Deligianni et al. (2014); first-episode psychosis datasets, University of Pittsburgh [VERIFY: approval ID] Salisbury et al. (2021a,b); Semmelweis dataset, Semmelweis University [VERIFY: approval ID] Racz et al. (2025); Parkinson's disease dataset, University of New Mexico [VERIFY: approval ID] Cavanagh (2021); Fribourg dataset, approval ID 054/13-CER-FR Padée et al. (2022). The analyses were carried out in accordance with the relevant guidelines and regulations and the Declaration of Helsinki.

A large language model (Claude, Anthropic) was used to assist with writing analysis code and drafting and editing the manuscript text. All analyses, results and text were reviewed and verified by the authors, who take full responsibility for the content. [VERIFY: adjust to the actual use]

4 Results

4.1 Classification

Band power separated Parkinson's disease from controls (AUC 0.786, 95% CI 0.63–0.92, $p = 0.010$) but not the psychosis or schizophrenia cohorts (Fig. 4, Table 2): FEP1 0.620 ($p = 0.12$), FEP2 0.646 ($p = 0.06$), FEP1 and FEP2 pooled ($n = 138$) 0.568 ($p = 0.14$), Semmelweis 0.571 ($p = 0.17$), Fribourg 0.539 ($p = 0.31$).

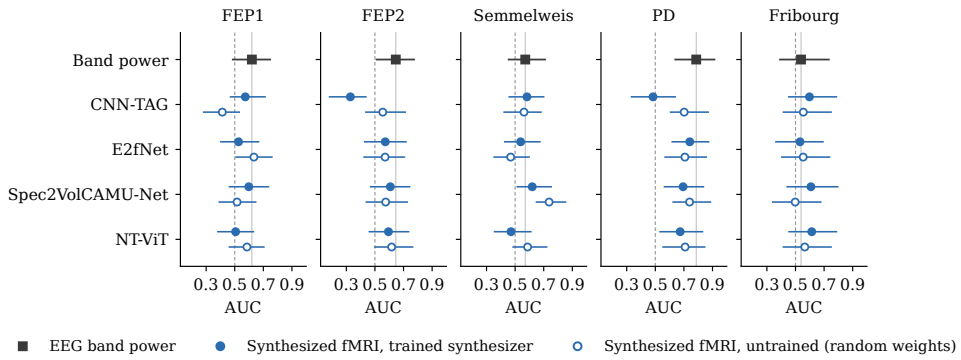


Fig. 4 Classification AUC by cohort and view. Mean AUC over 20 repeats of stratified 10-fold cross-validation with 95% bootstrap intervals, for EEG band power (squares) and the synthesized fMRI view of each synthesizer with learned weights (filled circles) or random weights (open circles). Dashed line: chance; grey line: band-power AUC. FEP1, FEP2: first-episode psychosis; Semmelweis: chronic schizophrenia; PD: Parkinson's disease; Fribourg: schizophrenia spectrum, task EEG.

In Parkinson's disease, where band power discriminates, synthesized views also discriminated but stayed below band power (E2fNet 0.741, $p = 0.01$; Spec2VolCAMU-Net 0.694, $p = 0.04$; NT-ViT 0.674, $p = 0.02$; CNN-TAG 0.484, $p = 0.54$). In the psychosis and schizophrenia cohorts, where band power did not discriminate, some synthesized views scored numerically above it (Semmelweis: Spec2VolCAMU-Net 0.620; Fribourg: 0.597–0.614 for CNN-TAG, Spec2VolCAMU-Net and NT-ViT), but only Spec2VolCAMU-Net in Semmelweis was significant ($p = 0.03$). Combining synthesized fMRI with band power changed AUC by -0.046 to $+0.112$. Five of the 20 uncorrected 95% intervals exclude zero (E2fNet and Spec2VolCAMU-Net in Semmelweis, Spec2VolCAMU-Net and NT-ViT in Parkinson's disease, E2fNet in Fribourg); under a normal approximation to the bootstrap distribution none reaches the Bonferroni threshold for 20 comparisons (largest $z \approx 2.7$, threshold 3.02). The untrained control gave a similar gain for three of the five, but not for E2fNet in Semmelweis ($+0.004$) or Spec2VolCAMU-Net in Parkinson's disease (-0.024).

Untrained synthesizers match trained ones. Where synthesized views discriminated, the same architectures with random weights did as well or better: in Parkinson's

Table 2 AUC per cohort (repeated stratified 10-fold, mean over 20 repeats; permutation p in parentheses). Trained / untrained: *fmri* view of the synthesizer with learned or random weights. Δ: $\mathrm{AUC}(v_0 \oplus v_2) - \mathrm{AUC}(v_0)$ with 95% paired bootstrap interval; uncorrected, 20 comparisons.

Cohort	View	Trained	Untrained	Δ trained	Δ untrained
FEP1	*bandpower*	0.620 (0.12)			
	CNN-TAG	0.574 (0.21)	0.413	−0.043 [−0.11, 0.03]	−0.088
	E2fNet	0.527 (0.31)	0.634	+0.005 [−0.08, 0.08]	−0.008
	Spec2VolCAMU-Net	0.598 (0.12)	0.516	−0.046 [−0.12, 0.02]	+0.006
	NT-ViT	0.506 (0.34)	0.585	−0.023 [−0.08, 0.03]	+0.009
FEP2	*bandpower*	0.646 (0.06)			
	CNN-TAG	0.328 (1.00)	0.555	−0.025 [−0.07, 0.02]	+0.028
	E2fNet	0.572 (0.22)	0.571	+0.000 [−0.04, 0.04]	−0.001
	Spec2VolCAMU-Net	0.608 (0.16)	0.575	+0.024 [−0.02, 0.07]	−0.001
	NT-ViT	0.595 (0.06)	0.617	+0.003 [−0.03, 0.03]	−0.003
Semmelweis	*bandpower*	0.571 (0.17)			
	CNN-TAG	0.583 (0.17)	0.562	+0.010 [−0.05, 0.07]	+0.071
	E2fNet	0.539 (0.29)	0.469	+0.065 [0.01, 0.12]	+0.004
	Spec2VolCAMU-Net	0.620 (0.03)	0.738	+0.112 [0.04, 0.20]	+0.168
	NT-ViT	0.471 (0.60)	0.587	−0.018 [−0.11, 0.08]	+0.094
PD	*bandpower*	**0.786** (0.01)			
	CNN-TAG	0.484 (0.54)	0.702	+0.019 [−0.00, 0.05]	+0.018
	E2fNet	0.741 (0.01)	0.707	+0.003 [−0.02, 0.03]	+0.032
	Spec2VolCAMU-Net	0.694 (0.04)	0.740	+0.035 [0.01, 0.07]	−0.024
	NT-ViT	0.674 (0.02)	0.708	+0.034 [0.01, 0.07]	+0.022
Fribourg	*bandpower*	0.539 (0.31)			
	CNN-TAG	0.597 (0.15)	0.555	+0.000 [−0.07, 0.06]	+0.004
	E2fNet	0.532 (0.31)	0.554	+0.075 [0.00, 0.15]	+0.061
	Spec2VolCAMU-Net	0.608 (0.11)	0.498	+0.009 [−0.05, 0.06]	+0.044
	NT-ViT	0.614 (0.12)	0.565	+0.007 [−0.03, 0.04]	+0.031

disease 0.702 (CNN-TAG), 0.707 (E2fNet), 0.740 (Spec2VolCAMU-Net) and 0.708 (NT-ViT), against 0.484, 0.741, 0.694 and 0.674 trained; in Semmelweis the untrained Spec2VolCAMU-Net reached 0.738 against 0.620 trained, and its fusion gain (+0.168, the only one beyond the Bonferroni threshold under the same approximation, $z \approx 4.0$) exceeded the trained one (+0.112).

Contrastive sinusoid separation. The contrastive classifier discriminated only in Parkinson's disease, at the smallest attainable p, and stayed below band power there (Table 3). Initialising the encoder from the NODDI-trained synthesizer did not improve on re-initialisation.

4.2 What the synthesizers learn

On NODDI, the EEG-free template reaches SSIM 0.676 and PSNR 21.2 under the protocol of Roos et al. (2025), at or above the published values of E2fNet (0.605), NT-ViT (0.581) and Spec2VolCAMU-Net (0.693). Retrained with trigger-aligned EEG,

Table 3 Contrastive sinusoid separation: AUC (repeated stratified 10-fold, 3 repeats), 95% bootstrap interval and permutation p (20 permutations; minimum 0.048).

Cohort	Re-initialised encoder	Pretrained encoder	*bandpower*
FEP1	0.485 [0.39, 0.67] (0.57)	0.599 [0.48, 0.73] (0.19)	0.620
FEP2	0.476 [0.34, 0.63] (0.52)	0.433 [0.27, 0.54] (0.76)	0.646
Semmelweis	0.538 [0.46, 0.72] (0.33)	0.562 [0.45, 0.70] (0.24)	0.571
PD	0.701 [0.58, 0.86] (0.048)	0.660 [0.53, 0.82] (0.048)	0.786
Fribourg	0.559 [0.38, 0.74] (0.33)	0.602 [0.44, 0.77] (0.14)	0.539

all synthesizers lost to or tied the template (SSIM 0.631–0.669 against 0.675; RMSE worse, $p \leq 0.02$), performed the same with EEG paired from the file start or with ECG alone, and showed no voxelwise temporal correlation beyond shifted or swapped EEG after detrending, motion regression and selection of steady-state volumes (three seeds; edge and interior voxels). With the fair target, EEG predicted a statistically detectable but negligible part of BOLD in held-out subjects: ridge from full spectra $r = 0.026$, UnEBOLT $r = 0.035$ (above shift and swap nulls, $p = 0.02$ and $p = 0.03$), with $R^2 \approx 0$ for both; within subjects, $r \approx 0$.

5 Discussion

Synthesized fMRI does not reliably improve EEG-based diagnosis in these cohorts. With four synthesizers, a contrastive fine-tuning method and five cohorts, synthesized views stayed below band power in the one cohort where EEG carried a clear signal, and their addition to band power gave small gains that did not survive correction for multiple comparisons and were, in three of five cases, reproduced by untrained networks of the same architecture. Where synthesized fMRI discriminated patients from controls, most clearly in Parkinson's disease, the discriminative content was what these networks extract from EEG spectra regardless of training. Fine-tuning the synthesizer's sinusoid space with the contrastive loss did not change this, and starting it from the learned mapping gave no advantage over starting from random weights.

Why. A synthesized volume can only carry information present in the EEG it is computed from, and the supporting analyses show how little of that information is haemodynamic. Under the image protocol used by published synthesizers, models reproduce the static anatomy that an EEG-free template already captures and ignore EEG timing; under a fair BOLD target, EEG predicts a statistically detectable but negligible share of the variance, even with a state-of-the-art model. The psychosis and schizophrenia cohorts add a second limit: resting band power itself discriminated only weakly there, so there was little EEG signal for any view to transform.

A checklist for diagnostic claims. Table 4 lists the controls of the protocol, the alternative explanation each rules out and what it showed here. Two of them concern synthesis itself: the NODDI dataset, the most used benchmark for volumetric synthesis, requires pairing EEG and fMRI through the scanner markers and selecting channels by name, two steps invisible to SSIM and PSNR, which an EEG-free template already maximises. We recommend that synthesis studies report an EEG-free template and temporal metrics against shift and swap nulls, and that diagnostic uses of synthesized fMRI report an EEG-only baseline, untrained-network controls and checks for acquisition differences between groups. A synthesizer that passes these controls would provide the first evidence that synthesized fMRI adds diagnostic information; the released code implements them and can be applied to new synthesizers.

5.1 Limitations

Sample sizes are modest (43–77 subjects per cohort), so AUC intervals are wide; null results bound but do not exclude small effects. Resting EEG carried weak signal in the psychosis and schizophrenia cohorts, so the evaluation is most informative in Parkinson's disease. The synthesizers were trained on one resting-state, eyes-open, 1.5 T dataset of 16 healthy adults; the clinical EEG differed in system, montage (interpolated), eye condition (Semmelweis eyes closed) and paradigm (Fribourg task), and no

Table 4 Controls for diagnostic claims from synthesized fMRI: what each rules out and what it showed in this study.

Control	Alternative explanation ruled out	In this study
EEG-only baseline under identical validation	Synthesized fMRI adds nothing beyond the EEG it is computed from	Band power above every synthesized view in Parkinson's disease
Fusion test, paired bootstrap, correction for multiplicity	Chance gains among many model–cohort comparisons	No trained gain survived correction for 20 comparisons
Untrained-network control	Discrimination by a fixed random projection of EEG spectra	Untrained synthesizers matched trained ones in Parkinson's disease (AUC 0.70–0.74)
Channel selection by type; acquisition checks	Group differences in recording set-up	Auxiliary-channel layout differed between groups in one release (Fribourg)
Subject-level nested cross-validation; permutation tests	Windows of one subject in training and test folds; tuning on test data; anti-learning bias of leave-one-out	All estimates subject-level with permutation p
EEG-free template image	Fidelity that reflects static anatomy	Template matched or exceeded all synthesizers (SSIM)
Temporal metrics against shift and swap nulls	Fidelity unrelated to EEG timing	Voxelwise $r \approx 0$; parcel $r \approx 0.03$ on a fair BOLD target
Trigger alignment and channels by name (NODDI)	EEG paired with the wrong volumes or electrodes	EEG starts 13–130 s before the first volume in 15 of 16 subjects

clinical cohort has paired fMRI, so synthesis fidelity in patients cannot be measured. Patients were medicated in the psychosis and schizophrenia cohorts, and Semmelweis patients were recorded earlier in the day. Each subject's synthesized sequence was summarised by its temporal mean and standard deviation; other summaries, such as synthesized connectivity, were not tested. The contrastive classifier used fixed hyperparameters and fewer permutations than the other views. CATD could not be evaluated, and NeuroBOLT/UnEBOLT were used for fidelity only.

6 Conclusions

We introduced a controlled protocol for testing whether EEG-synthesized fMRI aids diagnosis and applied it to four synthesizers and five public cohorts spanning first-episode psychosis, chronic schizophrenia and Parkinson's disease. Synthesized fMRI did not reliably improve classification over EEG band power (no gain survived

correction for multiple comparisons), untrained networks reproduced most of its discriminative content, and the contrastive sinusoid-separation method did not change this. These results set a baseline: claims that synthesized fMRI aids diagnosis should be tested against an EEG-only baseline and untrained-network controls, and synthesizers should first show fidelity beyond an EEG-free template and EEG-timing nulls. The protocol, code and checklist are released so that future synthesizers can be evaluated in the same way. An extension of the contrastive formulation to more than two classes is sketched in Additional file 1.

List of abbreviations

AUC: area under the receiver operating characteristic curve; BOLD: blood-oxygen-level-dependent; BPRS: Brief Psychiatric Rating Scale; CI: confidence interval; CNN-TAG: convolutional neural network with topographical attention graph; ECG: electrocardiogram; EEG: electroencephalography; EOG: electrooculogram; EPI: echo-planar imaging; FEP: first-episode psychosis; fMRI: functional magnetic resonance imaging; IAAFT: iterative amplitude-adjusted Fourier transform; ICD-10: International Classification of Diseases, 10th revision; LOOCV: leave-one-out cross-validation; MSE: mean squared error; PANSS: Positive and Negative Syndrome Scale; PCA: principal component analysis; PD: Parkinson's disease; PSNR: peak signal-to-noise ratio; RMSE: root mean squared error; SSIM: structural similarity index measure; STFT: short-time Fourier transform; TR: repetition time.

Supplementary information. Additional file 1 (PDF). Contrastive sinusoid separation: periodicity of the cosine projection (Fig. S1), the separation property with its proof (Proposition S1) and the extension to more than two classes (Fig. S2).

Declarations

Availability of data and materials. All datasets analysed in this study are publicly available: NODDI simultaneous EEG-fMRI (Open Science Framework, https://osf.io/94c5t) Deligianni et al. (2014); first-episode psychosis resting EEG, OpenNeuro ds003944 and ds003947 Salisbury et al. (2021a,b); Semmelweis schizophrenia resting EEG, Zenodo record 14808296 Racz et al. (2025); Parkinson's disease resting EEG, OpenNeuro ds003490 Cavanagh (2021); Fribourg Ultimatum Game EEG, OpenNeuro ds004000 Padée et al. (2022). The code is available at [VERIFY: repository URL and archived version (e.g. Zenodo DOI)]. Project name: [VERIFY: name]; operating system: Linux; programming language: Python; other requirements: PyTorch, MNE-Python, scikit-learn, FSL; licence: [VERIFY: licence].

Competing interests. The authors declare that they have no competing interests. [VERIFY: confirm]

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Authors' contributions. [VERIFY: e.g. DC designed the study, implemented the methods, performed the analyses and wrote the manuscript. RH supervised the study and revised the manuscript. All authors read and approved the final manuscript.]

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